INCH-POUND

MIL-S-50348A (AR)
31 August 1993
SUPERSEDING
MIL-S-50348 (MU)
10 November 1969

MILITARY SPECIFICATION

SYNCHRONIZER, PULSE: 10554894

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for a pulse synchronizer, used in Test Set, Subassembly MX-8048/TPM-22A, part of Radar Test Set AN/TPM-22A (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A **FSC** 4931

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SPECIFICATIONS

MILITARY

MIL-F-13926	- Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-T-28800	- Test Equipment for use with Electrical and Electronic Equipment, General Specification for
MIL-I-45607	- Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

MILITARY

MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the DODSSP - Customer Service, Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

10554527	- Test Set, Subassembly MX-8048/TPM-22A
10554601	- Pulse Synchronizer Schematic
10554894	- Pulse Synchronizer
12950877	- Requirements for Soldering Electrical and Electronic Assemblies

SPECIAL PACKAGING INSTRUCTION

PS 10554894	- Packaging of Pulse Synchronizer

(Copies of other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 10554894, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the fine pulse delay assembly shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.

b. Inorganic protective surface finishings.

c. Part identification and marking.

d. Workmanship.

3.3 Environmental service conditions.

3.3.1 Vibration. The pulse synchronizer shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as specified in MIL-T-28800 for Type I, Class 3, Style F test equipment.

3.3.2 Shock. The pulse synchronizer shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10%, 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude operating. The pulse synchronizer shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

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3.3.4 Temperature.

3.3.4.1 Operating. The pulse synchronizer shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between +32°F and +131°F.

3.3.4.2 Non-operating (storage). The pulse synchronizer shall not suffer permanent damage when subjected to the constant temperature extremes of -60°F for a period of 24 hours and +160°F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The pulse synchronizer shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram 10554601.

3.5 Performance. When supplied with inputs specified in Table I, the pulse synchronizer shall yield the outputs described in the following paragraphs.

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TABLE I. Input characteristics.

INPUT	CHARACTERISTIC	TERMINAL
+5 V	+5.0 ± 0.6 Vdc	P1-30, 31, 32 (+); P1-24, 25, 26 (RET)
$\overline{\text{PRF}}$	20 KHz negative pulse from not less than +3.0 Vdc to 0 Vdc and return, pulse width 0.6 µsec	P1-35
$\overline{\text{T1}}$	Negative pulse, same frequency as $\overline{\text{PRF}}$, from not less than +3.0 Vdc to 0 Vdc and return, pulse width 0.8 µsec, delayed 34.38 µsec from $\overline{\text{PRF}}$	P1-16
$\overline{\text{T3}}$	Negative pulse, same frequency as $\overline{\text{PRF}}$, from not less than +3.0 Vdc to 0 Vdc and return, pulse width 0.8 µsec, delayed 36.51 µsec from $\overline{\text{PRF}}$	P1-36
$\overline{\text{T5}}$	Negative pulse, same frequency as $\overline{\text{PRF}}$, from not less than +3.0 Vdc to 0 Vdc and return, pulse width 0.8 µsec, delayed 38.64 µsec from $\overline{\text{PRF}}$	P1-8
$\overline{\text{T11}}$	Negative pulse, same frequency as $\overline{\text{PRF}}$, from not less than +3.0 Vdc to 0 Vdc and return, pulse width 0.8 µsec, delayed 45.02 µsec from $\overline{\text{PRF}}$	P1-20
$\overline{\text{STROBE}}$	Train of 15 negative pulses from not less than +3.0 Vdc to 0 Vdc and return, each 0.133 µsec wide on 1.064 µsec centers; leading edge of first pulse delayed 34.78 µsec from $\overline{\text{PRF}}$	P1-12
$\overline{\text{FINE DELAY}}$	Negative pulse, same frequency as $\overline{\text{PRF}}$, from not less than 3.0 Vdc to 0 Vdc and return, pulse width 0.21 µsec	P1-51

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3.5.1 Trigger outputs. When supplied with the +5 V, $\overline{T3}$, T5, and STROBE inputs of Table I, the pulse synchronizer shall yield the outputs of Table II.

TABLE II. Trigger outputs.

OUTPUT	CHARACTERISTIC	TERMINALS	TERMINATION
E/L TRIGGER	Positive pulse, same frequency as PRF, pulse width 0.35 µsec, pulse amplitude not less than 2.4 Vdc delayed 38.67 µsec from PRF, leading edge approximately coincident with leading edge of T5, into 150 ohms	P1-1,2	150 ohms
PRERANGE TRIGGER	Positive pulse, same frequency as PRF, pulse width 0.46 µsec, pulse amplitude not less than 2.4 Vdc, delayed 36.93 µsec from PRF, leading edge approximately coincident with leading edge of T3, into 150 ohms	P1-6,9	150 ohms

3.5.2 COARSE DELAY output. When supplied with the +5 V, PRF, T1, and the T11 inputs of Table I, and with exactly two terminals from among P1-18, 22, and 33 grounded the pulse synchronizer shall yield the outputs described in Table III at P1-49, when a load of 150 ohms is applied to that terminal.

TABLE III. Coarse delay output.

TERMINALS GROUNDED	OUTPUT
P1-18, 22	Positive pulse, same frequency as PRF, pulse width, 0.6 µsec, pulse amplitude not less than 2.4 Vdc, delayed 30 ns from PRF.
P1-18, 33	Positive pulse, same frequency as PRF, pulse width, 0.8 µsec, pulse amplitude not less than 2.4 Vdc, delayed 34.41 µsec from PRF.
P1-22, 33	Positive pulse, same frequency as PRF, pulse width, 0.8 µsec, pulse amplitude not less than 2.4 Vdc, delayed 45.05 µsec from PRF.

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3.5.3 Pulse outputs. When supplied with the +5 V, and $\overline{\text{FINE DELAY}}$ inputs described in Table I, the pulse synchronizer shall yield the outputs specified in Table IV.

TABLE IV. Pulse outputs.

OUTPUT	CHARACTERISTIC	TERMINALS	TERMINATION
VIDEO PULSE	Positive pulse, same frequency as PRF, pulse width 0.2 µsec, pulse amplitude not less than 2.4 Vdc, delayed 30 ns from FINE DELAY.	P1-17	150 ohms
$\overline{\text{TARGET PULSE}}$	Negative pulse, same frequency as PRF, pulse width 0.2 µsec, from not less than +2.9 Vdc to 0 Vdc and return, delayed 20 ns from $\overline{\text{FINE DELAY}}$.	P1-34	150 ohms

3.6 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Soldering. All soldering shall be in accordance with the requirements of Note 2 of Drawing 12950877.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-F-13926 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50°F to +90°F
b.	Relative humidity	90% RH Max.
c.	Pressure	28 to 32 inches Hg

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4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated.

Name	Drawing	Quantity
Pulse Synchronizer	10554894	3

4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

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4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies.

a. The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawings and specifications utilizing as a minimum the conformance criteria specified in this document and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

b. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.3). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provisions(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

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4.4.2.3 Functional tests. The tests in Table V shall be performed on a 100% basis.

TABLE I. Functional tests (100 percent inspection).

Characteristic	Requirement	Test Procedures
Critical	None	N/A
Major 101. Electrical 102. Performance	 3.4 3.5	 4.7.2 4.7.3
Minor	None	N/A

4.5 Special sampling. One pulse synchronizer out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests in Table V and shall then meet the requirements and tests in Table VI. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE VI. Special sampling.

Characteristic	Requirement	Test Procedures
Critical	None	N/A
Major 103. Vibration 104. Shock 105. Altitude 106. Temperature 107. Humidity (nonoperating)	 3.3.1 3.3.2 3.3.3 3.3.4 3.3.5	 4.7.1.1 4.7.1.2 4.7.1.3 4.7.1.4 4.7.1.5
Minor	None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test, as well as to current production has been accomplished in accordance with the provisions of the contract.

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4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 wherever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the standard test equipment listed in Table VII (Test equipment list) shall be utilized in the performance of the applicable test methods and procedures as specified in 4.7.

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TABLE III. Test equipment list.

Item Number	Description
1. Power supply	5.0 ± 0.3 Vdc, regulated; ac ripple less than 50 mV peak-to-peak; capacity 300 mA.
2. Pulse generator	20 KHz repetition rate. Negative-going pulses from +3 Vdc level to 0 Vdc and return, into 150 ohms. Pulse width 0.2 - 0.8 µsec. Must have trigger output compatible with external drive requirements of items 3 and 4. Trigger out must be coincident with output pulse.
3. Pulse generator	20 KHz repetition rate. Negative-going pulses from +3 Vdc level to 0 Vdc and return. Pulse width 0.8 µsec. Must have external drive input compatible with trigger output capability of item 2. Output must be capable of being delayed 2.0 - 0.3 µsec from external drive input.
4. Timing unit	Input drive requirement must be compatible with trigger output capability of item 2. Capable of generating output gate 20 ns to 10 µsec wide, 20 ns to 10 µsec after leading edge of input trigger.
5. Pulse generator	0.9 - 1.0 MHz repetition rate. Negative-going pulses from +3 Vdc level to 0 Vdc and return into 150 ohms. Pulse width 133 ns. Must be gated by output gate of item 4. Leading edge of first output pulse must be coincident with leading edge of input gate.
6. Oscilloscope	Dual channel, vertical passband DC to 15 MHz. Rise time of scope plus measuring probe 10 nanoseconds or less. Sweep period 50 nanoseconds per centimeter or less. Input capacitance less than 25 picofarads.
7. Test harness	Per Figure 1 and Table VIII.

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TABLE VIII. Parts list for test harness (Table VII, item 7).

DESIG.	PART	DESCRIPTION
P1	Connector	Malco 5893232-1, Elco 07-5422-0198-01, or equivalent
S1	Switch	SPST
S2	Switch	SPST
S3	Switch	SPST
TB1	Terminal strip	9 terminals
TB2	Terminal strip	5 terminals

NOTE: Wire, mounting hardware, 150 ohms 1/4 w 10% resistors, per Figure 1.

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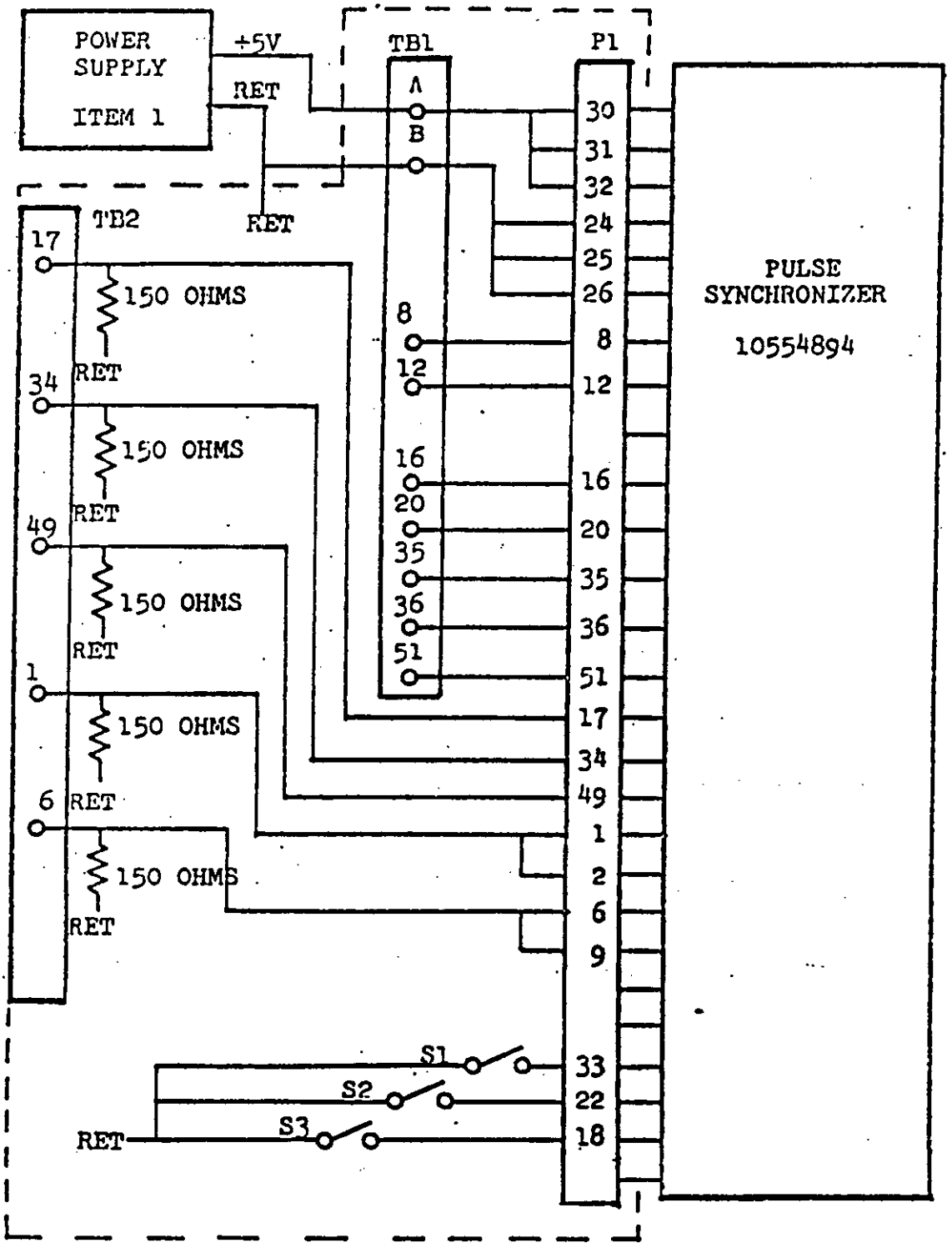


FIGURE 1. Test harness connections.

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4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. This test shall be performed as specified in MIL-T-28800 for Type I, Class 3, Style F test equipment with the exception of Step 4.c Resonance Dwell does not apply. The pulse synchronizer shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the pulse synchronizer shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz.

After the application of all the shock impulses, the shock synchronizer shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating). This test shall be performed as specified in Procedures II, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

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4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.7.3 Performance. Adjust power supply (Table VII, item 1) to delivery +5 V input of Table I, and connect to test harness (Table VII, item 7) as shown in Figure 1.

4.7.3.1 Trigger outputs. Adjust pulse generator (Table VII, item 2) to deliver 20 KHz pulse, from not less than +3.0 Vdc to 0 Vdc and return, pulse width 0.8 µsec and connect as shown in Figure 2. Connect trigger output of pulse generator (Table VII, item 2) to external drive (input) of pulse generator (Table VII, item 3). Adjust pulse generator (Table VII, item 3) to deliver pulses delayed 2.13 µsec (leading edge to leading edge) from external drive; pulse amplitude from not less than +3.0 Vdc to 0 Vdc and return, pulse width 0.8 µsec. Connect output of pulse generator (Table VII, item 3) as shown in Figure 2. Connect timing unit (Table VII, item 4) to trigger output of pulse generator (Table VII, item 2). Adjust timing unit to deliver gate 3.5 µsec wide, with leading edge delayed 0.4 µsec from leading edge of input drive. Connect output (gate) to gate input of pulse generator (Table VII, item 5). Adjust pulse generator (Table VII, item 5) to deliver pulses at 940 KHz, from not less than +3.0 Vdc to 0 Vdc and return, pulse width 133 nsec, and

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connect as shown in Figure 2. Use oscilloscope (Table VII, item 6) to check for presence of simulated E/L TRIGGER and PRERANGE TRIGGER outputs of Table II at test harness terminals TB2-1 and 6, respectively. Timing of signals for this test is shown in Figure 3.

4.7.3.2 COURSE DELAY output. Set up pulse generator (Table VII, item 2) as in 4.7.3.1 and connect output to test hardness TB1-35. Close switches S2 and S3 of test harness. Leave switch S1 open. Use oscilloscope (Table VII, item 6) to check for presence of positive pulse, not less than 2.4 Vdc, same width as output pulse of pulse generator, and coincident with it, at test harness TB2-49. Repeat with pulse generator output connected to test harness TB1-16, S1 and S3 closed, and S2 open. Repeat again with pulse generator output connected to test harness TB1-20, S1 and S2 closed, and S3 open.

4.7.3.3 Pulse outputs. Adjust pulse generator (Table VII, item 2) to delivery FINE DELAY signal of Table I, and connect to test harness TB1-51. Use oscilloscope (Table VII, item 6) to check for presence of VIDEO PULSE and TARGET PULSE of Table IV at test harness terminals TB2-17 and 34, respectively.

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of Note 2 of Drawing 12950877 as specified in paragraph 3.7.

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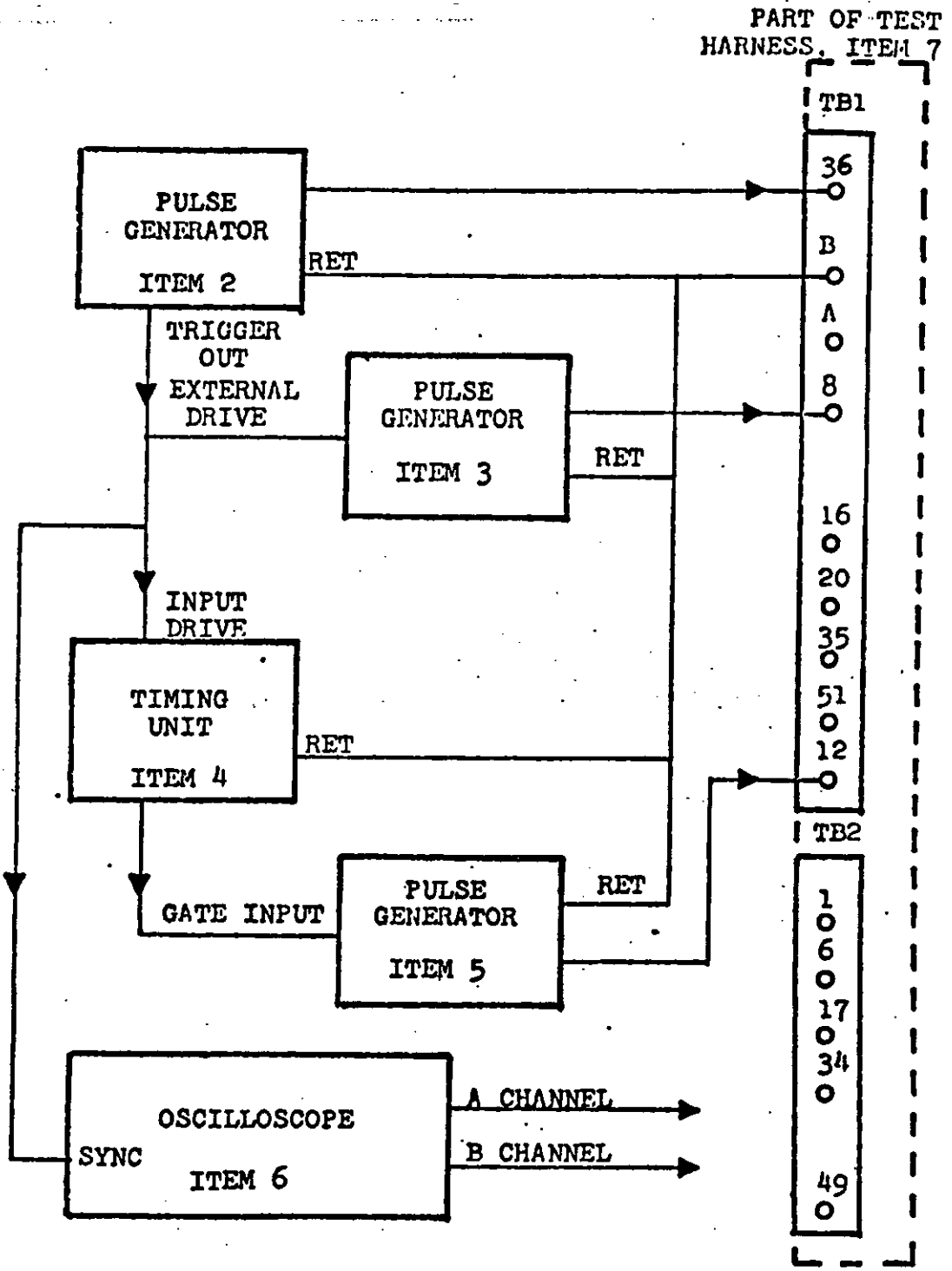
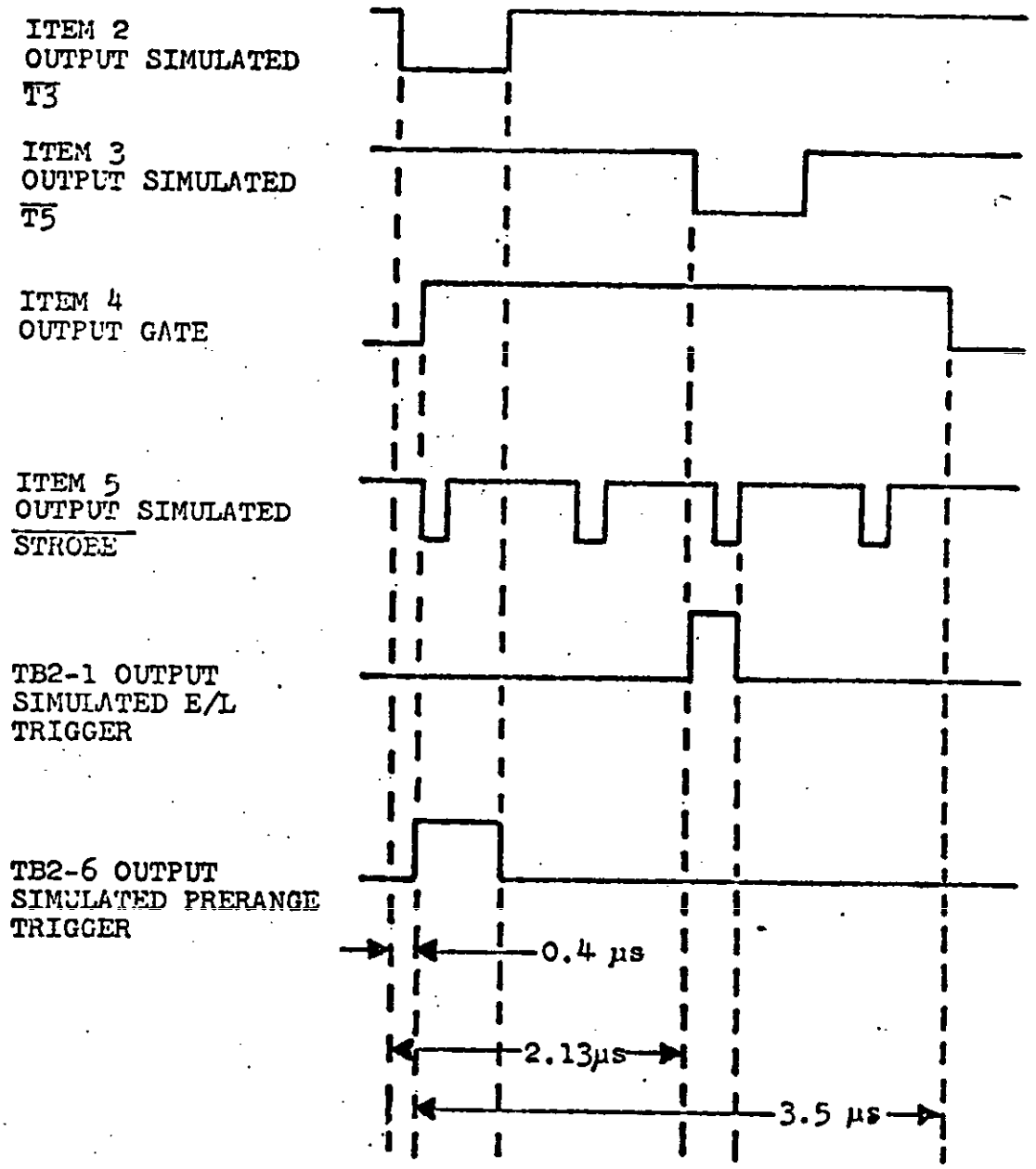


FIGURE 2. Trigger outputs check.

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NOTE: PULSES SHOWN SQUARE FOR CONVENIENCE ONLY

FIGURE 3. Timing for trigger outputs.

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5. PACKAGING

5.1 General. Preservation, packing and marking shall be in accordance with Special Packaging Instruction PS 10554894. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The pulse synchronizer is used in Test Set, Subassembly MX-8048/TPM-22A, part of Radar Test Set AN/TPM-22A. The pulse synchronizer develops certain pulse outputs needed elsewhere in, and as outputs from, Test Set, Subassembly MX-8048/TPM-22A.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification

b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.1.1).

c. Requirements for submission of first article sample.

d. Applicable stock number.

e. Packaging requirements, if other than specified in Section 5.

f. Serialization requirements, if applicable.

g. Certificate of conformance for each lot or shipment of product.

6.3 Submission by the contractor.

a. Inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

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b. Alternative quality conformance provisions for approval. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation to the technical activity responsible for the preparation of this specification.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (key word) listing.

Fire Control System
Radar Test Set AN/TPM-22A
Test Set, Subassembly MX-8048/TPM-22A
VADS

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 4931-A264)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
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4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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